

Comp 442 Group Assignment 1

“Phish Detector”

Members: Bachviet Nguyen, Carlos Bautista, Finn Bishop, Haley Kenney, Patrick Banting

Project Overview

The goal of this project is to develop a machine learning-based phishing and spam email detection system. We will use a publicly available spam dataset from Kaggle containing labeled messages classified as spam or ham. The purpose of this assignment is to understand the dataset, identify its structure and feature types, and determine how the data can be prepared for machine learning and natural language processing tasks.

Dataset Acquisition

Dataset Name: Email Spam Detection Dataset

Source: Kaggle

Location: <https://www.kaggle.com/datasets/abdmental01/email-spam-dedection>

Method of Acquisition:

- Downloaded from Kaggle as a CSV file.
- Imported into the project environment for analysis and preprocessing.

Problems Encountered:

- Needed to verify dataset structure and labels.
- Potential missing values, duplicate records, and class imbalance.

Resolution:

- Data cleaning and preprocessing will be performed before model training

Dataset Description

Dataset Format:

- CSV (Comma-Separated Values)

Dataset Size:

- 5,572 records
- 2 attributes

Fields:

- Category: Indicates whether a message is Spam or Ham.
- Message: Contains the text content of the message.

Data Characteristics:

- Labeled dataset for supervised learning.
- Message text will be converted into numerical features for machine learning.

Dataset Structure

Structured Data: The dataset is organized in rows and columns within a CSV file.

Semi-Structured Data: Not present in this dataset.

Unstructured Data: The Message field contains raw text that requires NLP preprocessing.

The dataset combines structured tabular data with unstructured text data.

Dataset Types

(tabular data, text data, and other applicable categories if any)

Feature Types

Category/Label Field

- Identifies whether an email is spam or legitimate/ham
- Used as the target variable for supervised learning

Message Field

- Contains the raw email text/content
- Main input used for text classification

Text-Based Features

- Important words and phrases from data content
- Extracted using TF-IDF and n-grams

Metadata/Engineered Features

- Message length
- Word count
- Digit count
- Suspicious keyword count
- URL/link indicators

Feature Classification

- Target feature: spam/ham label
- Descriptive features: message text and extracted text patterns

Data Exploration & Sample Records

- Dataset contains labeled email messages for spam classification
- Each record represents one email/message sample

Main fields expected:

- Label/category: spam or legitimate
- Message/text: email content

Basic observations:

- Spam messages may contain urgent wording, links, money-related terms, or promotional language
- Legitimate messages are expected to contain more normal communication patterns

Initial exploration will include:

- Dataset size: [insert number of rows after loading]
- Number of spam vs legitimate emails
- Missing or empty messages
- Duplicate records
- Average message length

Text Feature Extraction

- Clean raw email text before modeling
- Convert all text to lowercase
- Remove extra spaces, symbols, and unnecessary characters
- Tokenize messages into individual words or terms
- Remove common stopwords when useful

Apply TF-IDF vectorization

- Converts email text into numerical features
- Gives higher weight to important words

Use n-grams

- Captures phrases such as “click here,” “verify account,” or “limited offer”

Add engineered text features:

- Message length
- Word count
- Number of digits
- Number of URLs
- Suspicious keyword count

Extracted Features

(message length, word count, url count, suspicious keywords, tf-idf vectors)

Challenges & Observations

(dataset limitations, potential class imbalance, noise in text data, preprocessing considerations)

Conclusion

(summary of findings, how the dataset supports the project)